A self-certifying simplex for rigorous polytope-hull contraction in interval branch and bound

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Abstract

Polytope-hull contractors bound each variable of an interval branch-and-bound node over a linear relaxation by solving $2n$ linear programs. Current implementations delegate them to a floating-point LP solver and recover rigor afterwards from its dual solution. We present a self-certifying simplex (SC) that computes its own multipliers: writing the relaxation as $\bar{A}z = 0$ with interval-bounded slacks, any multiplier vector yields a certificate valid on the box and every sub-box, and SC is designed around it, with a free dual-feasible start for every bound, one tableau shared by the $2n$ problems, a warm start chosen per program and a hybrid tableau/revised engine. Replaying the same polytope-hull passes inside the same contractor, a pass of SC costs 0.47 times that of SoPlex on optimization problems and 0.55 times on systems of equations, with the same contraction. SC does not pivot less: refactorizing at every call, verifying the solution and presolving each relaxation account for

two thirds of the difference, which SC avoids by sharing one tableau among the $2n$ problems. The whole solver is 17–20% faster on 240 global optimization problems and 9–10% faster on 242 systems of equations, with slightly fewer cells, and SC never exceeds the LP hull beyond the tolerance of an LP oracle.

1 Introduction

Interval branch-and-bound solvers compute, with mathematical certainty, an enclosure of all solutions of a system of nonlinear equations and inequalities, or the global optimum of a constrained problem [22]. Their efficiency depends on *contractors*, operators that shrink the current box without losing solutions [7]. Among them, contractors based on linear relaxations are among the most effective: the constraints are linearized over the box, by convex interval Taylor forms [3] or affine arithmetic [25, 2], and each variable is minimized and maximized over the resulting polytope. In Ibex this is the polytope-hull contractor, the linear part of X-Newton, which we introduced in [3]; we later combined the affine and Taylor-based relaxations [2].

Each call solves up to $2n$ linear programs, and the rigor of the result cannot come from the LP solver, which works in floating point. Current implementations therefore treat the LP solver as a black box and certify its output a posteriori: the dual solution is turned into a safe bound by the Neumaier–Shcherbina procedure [23], and an infeasibility claim is accepted only if the returned Farkas ray passes the same verification. This design has a cost that is easy to miss. When the verification fails, which happens when the dual is inaccurate or the solver works in an unfavourable representation, the bound is lost; when the solver reports infeasibility without a

verifiable ray, the pass is abandoned and the node survives. On our benchmark, these losses, and not the quality of the relaxation, explain most of the pruning gap between contractors that share the same polytope (Section 6). The design has a second cost, which turns out to be the larger one: the $2n$ LPs of a node are small and closely related, typically two or three pivots apart, and each is a separate call to a general-purpose solver, whose work per call then exceeds the pivots themselves (Section 6).

Positioning. The certificate we use is not new. That *any* multiplier vector yields a valid bound once the residual is evaluated in interval arithmetic is the principle of Neumaier and Shcherbina [23]. Domes and Neumaier [10] applied it to filtering with linear relaxations: they solve the bound LPs with an external solver, collect the multipliers, and solve the preconditioned interval system row by row. In previous work [26] we wrote the relaxation in the homogeneous form used here, with one auxiliary variable per row, and showed that the optimal preconditioner for an interval Gauss–Seidel projection is given by the dual solutions of the bound LPs; the equivalent valid inequalities appear in the optimization-based bound tightening of Gleixner et al. [13], and both works reuse them in sub-boxes. The present paper continues that line: there the multipliers were still computed by an external LP solver, and the preconditioner was recomputed only periodically to amortize its cost. On the solver side, Yamamura and coworkers [29, 30] observed that, after a bisection, the optimal tableau of the parent remains dual feasible for the child, so that an LP emptiness test can be resumed by the dual simplex method with few or no pivots per box.

What changes with a self-certifying simplex is *where the multipliers come from and how often a usable certificate is available.* In the approaches above, multipliers are a by-product of an exter-

nal solver and certification is a post-processing step; its cost and fragility lead to applying it to a few promising variables [10], to recomputing it only periodically and reusing it in between [26], or to abandoning the pass when it fails (Ibex). The dual-simplex warm start of Yamamura et al. concerns a single feasibility LP on a fixed relaxation, without rigorous certification. The self-certifying simplex is designed for the certificate: in the homogeneous form every basis can be made dual feasible by relocating nonbasic variables, so each of the $2n$ bound problems, and each problem after the box contracts, starts dual feasible; the $2n$ problems share one tableau; and the multipliers of the current basis give a valid certificate at every bound and, with the dual method, at every iteration. Bound tightening by optimization is also central in MINLP solvers [5, 13], where rigor is not required; the heuristic of skipping bounds already attained by a primal point, used by Ibex, comes from Baharev, Achterberg and Rév [4].

Contributions.

- The use, in the homogeneous form of the relaxation, of a standard property of boxed LPs [18]: every basis becomes dual feasible for every bound problem by relocating its nonbasic variables, so that the $2n$ problems, and each problem after the box contracts, share one tableau and start without a phase 1 (Section 5).

- A specialized simplex for the $2n$ problems: shared tableau, long-step dual ratio test, devex pricing, box-scaled tolerances, warm starts between bounds chosen by how many nonbasic variables change bound (dual on systems of equations, primal on optimization problems), and a hybrid tableau/revised engine (Section 4).

- Properties of the certificate: validity for any multipliers and any sub-box, exactness at an optimal basis, and, under the dual method, a valid lower bound that is nondecreasing along the iterations (Section 5).

- A controlled evaluation in Ibex, on 242 systems of equations and 240 optimization problems, in which the same contractor runs either SC or a black-box LP solver on the same LPs, feeding the same certificate; a rigor check against the LP hull; a bank of polytope-hull passes replayed outside the search, where a pass of SC costs 0.47 (optimization) and 0.55 (systems) times that of the black box with the same contraction, and an attribution of that difference to the parts of a call of the black box; an analysis of the LPs where the black box pivots less; and an ablation (Section 6).

2 Background and related work

Interval solvers and linear relaxations. Interval branch and bound alternates bisection with contraction. Ibex, on which we build, composes constraint propagation (HC4 [6]), shaving-based consistencies such as ACID [24], the interval Newton operator for square systems [21], and contractors based on linear relaxations [7]. The polytope-hull contractor computes a linear relaxation of the constraints over the box and bounds each variable over it; in X-Newton [3] the relaxation comes from convex interval Taylor forms, and the affine arithmetic relaxation of Ninin et al. [25] is an alternative which we later combined with it [2]. QUAD [19] and GloptLab [10] use linear relaxations in the same way. In previous work [26] we wrote the relaxation in the homogeneous form used here and showed that the optimal preconditioner for a Gauss–Seidel

projection over it is given by the dual solutions of the bound LPs. Yamamura and coworkers [29, 30] solve an LP emptiness test at each box of a bisection search and resume it by the dual simplex method from the parent's tableau.

Rigorous bounds for linear programs. A floating-point LP solver returns approximate solutions, and two lines of work make its results rigorous. The first certifies them a posteriori. Neumaier and Shcherbina [23] showed that any approximate dual vector yields a safe bound once the residual is evaluated in interval arithmetic, and that an approximate Farkas ray proves infeasibility under the same verification; Jansson [16] gave rigorous lower and upper bounds based on the same principle, implemented in Lurupa [17]. In filtering, Domes and Neumaier [10] apply this certificate to the bound LPs of a linear relaxation, collecting the multipliers from an external solver for a few promising variables. The second line solves the LP exactly, in rational arithmetic [1], or to arbitrary precision by iterative refinement [14], as needed in exact mixed-integer programming [8]. Both treat the LP solver as given. The present work keeps the a posteriori certificate, which costs no more than one matrix–vector product, and changes the solver instead.

Optimization-based bound tightening. Minimizing and maximizing each variable over a relaxation is also a standard bound tightening technique in global optimization, where rigor is not required: range reduction in BARON [27], and optimization-based bound tightening in Couenne [5] and SCIP. Gleixner et al. [13] make it cheaper by filtering bounds already attained by feasible points, by ordering the LPs greedily, and by turning the dual solutions into valid inequalities that are propagated in the subtree. The skipping rule used by Ibex comes from Baharev, Achterberg and Rév [4]: a

bound attained by a primal point found earlier needs no LP, and the next LP is warm-started from the previous one.

Simplex techniques. Our simplex uses standard ingredients of modern implementations, described in detail by Koberstein [18] for the dual method: the bounded-variable dual simplex with the long-step (bound flipping) ratio test [20], Harris' ratio test and devex pricing [15], an approximation of steepest edge [11], and the revised method with a sparse LU factorization of the basis updated by Forrest and Tomlin's method [12], as in SoPlex [28]. What is specific to our setting is not any of these techniques but their use for $2n$ closely related LPs over the same constraints, with a certificate attached to every basis.

3 The homogeneous relaxation and its certificate

Let $\mathbf{x} \in \mathbb{IR}^n$ be the current box. A linearizer returns a matrix $A \in \mathbb{R}^{m\times n}$ of floating-point numbers and an interval vector $\mathbf{b} \in \mathbb{IR}^m$, possibly with infinite bounds, such that

$$x \in \mathbf{x} \text{ is a solution } \implies Ax \in \mathbf{b}. \tag{1}$$

We introduce one slack per row, $s = Ax$, and write the relaxation in the homogeneous form

$$\bar{A}z = 0, \qquad z = (x, s) \in \mathbf{z} := \mathbf{x} \times \mathbf{s}, \qquad \bar{A} := [\, A \;\; -I_m \,], \tag{2}$$

where $\mathbf{s}_i := \mathbf{b}_i \cap \square(A_{i:}\,\mathbf{x})$ and $\square(\cdot)$ denotes evaluation in interval arithmetic. We call $P(\mathbf{x}) := \{z \in \mathbf{z} : \bar{A}z = 0\}$ the *relaxation polytope*.

Lemma 1. *If $x^* \in \mathbf{x}$ is a solution, then $(x^*, Ax^*) \in P(\mathbf{x})$.*

Proof. $A_{i:}x^* \in \mathbf{b}_i$ by (1) and $A_{i:}x^* \in \square(A_{i:}\mathbf{x})$ by the inclusion property of interval arithmetic; the equation holds by construction of s. □

Definition 2 (Certificate)**.** For $\lambda \in \mathbb{R}^m$, let $\gamma \in \mathbb{IR}^{n+m}$ be an interval vector that contains $\gamma := \bar{A}^T\lambda$, computed in outward-rounded interval arithmetic. For $k \in \{1, \ldots, n+m\}$ with $0 \notin \gamma_k$, the *certified enclosure* of z_k is

$$\mathbf{C}_k(\lambda, \mathbf{z}) := -\Big(\textstyle\sum_{j \neq k} \gamma_j\, \mathbf{z}_j\Big) \Big/ \gamma_k,$$

and the *residual* is $\mathbf{R}(\lambda, \mathbf{z}) := \sum_j \gamma_j\, \mathbf{z}_j$.

A contraction step replaces $\mathbf{z}_k$ by $\mathbf{z}_k \cap \mathbf{C}_k(\lambda, \mathbf{z})$; an emptiness test checks $0 \notin \mathbf{R}(\lambda, \mathbf{z})$. Section 4 explains how the simplex produces λ, and Section 5 why these steps are valid and when they are tight.

Why not a rigorous simplex. Validity does not depend on how λ was computed: any multiplier vector gives a valid bound, and a poor one only gives a weaker bound. Rigor is therefore needed only in the final evaluation of γ and $\mathbf{C}_k$, not in the pivoting. Carrying intervals through the pivots instead pays the overestimation of interval Gaussian elimination [21], whose widths grow with the number of pivots, and solving each LP exactly in rational arithmetic, or to high precision by iterative refinement [1, 14], buys an exactness that the bound does not need, at a cost that $2n$ LPs per node cannot afford. An early version of our contractor pivoted in interval arithmetic: on a one-box benchmark it had to rebuild the tableau from the original relaxation 731 times to control the width

growth, and pivoting in floating point with certification of the final multipliers only was 2.8 times faster with essentially the same contraction (mean 24.0% against 24.6%).

4 A simplex for the $2n$ bound problems

For each structural variable x_k the contractor needs the two bounds of

$$\mathrm{LP}_k: \quad \min\ z_k \quad \text{s.t.} \quad \bar{A}z = 0,\ z \in \mathbf{z}, \tag{3}$$

and of its maximization, obtained with $-z_k$. A basis is a set of m columns of $\bar{A}$ forming a nonsingular matrix B; with cost vector $c = \pm e_k$, the multipliers are $\lambda^T = c_B^T B^{-1}$ and the reduced costs $d = c - \bar{A}^T\lambda$. For the bounded-variable simplex, a basis with each nonbasic variable at one of its bounds is *dual feasible* if $d_j \geq 0$ for nonbasic variables at their lower bound and $d_j \leq 0$ for those at their upper bound, and *primal feasible* if the basic variables lie within their bounds. This section describes how the simplex solves the $2n$ problems of a node; the design choices are evaluated in Section 6.

4.1 One pass over the bounds

At each node the relaxation is computed, written in the form (2), and loaded once. The affine linearizer writes an equation as two one-sided rows with the same normal; in the homogeneous form they are merged into one row whose slack has the interval of both sides, which halves m on square systems. The contractor then makes one pass over the up to $2n$ bounds with a single simplex: each LP starts from the final basis of the previous one. Every bound is contracted as soon as it is certified, so the next LP already sees the smaller box.

Two rules, both from the polytope-hull contractor of Ibex [4], decide which LPs are solved. A bound is *skipped* when a primal point obtained earlier in the pass already attains it, since its LP would return that value. Among the remaining bounds, the next one is the farthest from the last primal point. The tableau suggests a better order: for x_k basic in row r, the number of nonbasic columns whose reduced cost for the objective $\pm z_k$ would have the wrong sign is the number of primal infeasibilities to repair, and can be counted in $O(\text{nnz}(\text{row } r))$ without pivoting. Ordering by this count saves a few percent of CPU time (Section 6). The count predicts the work of the dual warm start; with the primal warm start of the optimizer (below) the previous vertex is kept, and the nearest bound first, whose optimum lies closest to it, needs fewer pivots: SC orders by the count in the solver of systems of equations and takes the nearest bound first in the optimizer. The black box has no tableau and can only order by distance; we give it the better of the farthest and the nearest bound first, and also compare both engines with the farthest-bound rule.

If the residual $\mathbf{R}(\lambda, \mathbf{z})$ of an LP excludes zero, the node is empty and the pass stops. If an LP ends without an optimal basis (iteration limit or numerical failure), its bound is not contracted and the pass goes on.

4.2 Starting and warm-starting

The first LP of a pass has no previous basis. It starts from the slack basis $B = -I_m$, which is dual feasible for every LP_k (Proposition 8), and is solved by the dual method, with no phase 1.

Between two consecutive LPs of the pass the objective changes, and the contracted bound changes the box. Two warm starts are possible from the previous optimal basis:

- *dual*: relocate every nonbasic variable to the bound indicated by the sign of its new reduced cost, z_k included, which restores dual feasibility (Proposition 8), and continue with the dual method; the certified bound is then available, and nondecreasing, at every iteration (Proposition 9);
- *primal*: keep the nonbasic variables where they are; the basis remains primal feasible, up to the contracted bound, and is reoptimized with primal steps, which is the usual warm start of an LP solver.

Both reach an optimal basis, so the certificate at termination is exact (Proposition 6) and the pruning is the same; whether the node is empty is decided in the first LP, common to both. They differ only in cost, and which one is cheaper depends on how the optima of consecutive bounds are related. An optimal vertex is given by a basis and by the bound on which each nonbasic variable rests. Table 1 compares the optima of consecutive bounds over the benchmarks of Section 6: the basis changes by 10–17% in both programs, but on systems of equations a median of 1.9 nonbasic variables per LP move to their opposite bound, against 0.3 on optimization problems. The dual warm start moves them for free, since relocation is not a pivot; the primal warm start pays one iteration for each, and on systems of equations its iterations exceed those of the dual by about the number of side changes (a median difference of 1.9 per LP, for 1.9 side changes); on optimization problems, with the nearest bound first, it even needs fewer iterations than the dual (2.1 against 2.5). The two warm starts thus have different costs: the primal one pays one iteration per side change, the dual one a fixed cost per LP, since relocation changes the values of the nonbasic variables and the basic values must be recomputed from the factorization. On systems of equations side changes dominate: the primal warm start makes 5.1 iterations per LP against 3.1, and on

Table 1: Optima of consecutive bounds of a pass (seed 1, 60 s per run, 165 systems and 141 optimization problems with at least 1 000 warm LPs in both warm starts). Medians, with quartiles in parentheses, of the rows of the relaxation, the basic variables that change, the nonbasic variables that move to their opposite bound, and the iterations per warm LP with each warm start.

		basis	side changes		iterations per LP	
benchmark	rows	changed	per LP	% nonbasic	dual	primal
systems	18 (10–32)	10%	1.9 (1.1–2.8)	16%	3.1	5.1
optimization	9 (5–17)	17%	0.3 (0.2–0.5)	4%	2.5	2.1

Bratu-0120 an LP costs about twice as much (422 against 223 μs, of which 224 μs in primal steps). On optimization problems few sides change, and the fixed cost of relocation dominates: on launch and ex5_3_2 (about 40 rows), an LP costs 143 against 158 and 128 against 168 μs with the primal warm start, which pivots slightly less and recomputes the basic values for 6–7 instead of 16–17 μs per LP; on ex2_1_9, whose LPs need one or two pivots, the primal warm start pivots 25% more and its LPs are still 6% cheaper. The side changes differ because of the rows. The box bounds of the variables change side in both programs (14% and 8% of the nonbasic variables); the slacks of the rows change side in 14% of the cases on systems of equations and in 0.3% on optimization problems. On systems every row comes from an equation and is a narrow slab: the interval of its slack is, on average, 13% of the range of the row over the box, so that the optima of consecutive bounds often rest on opposite faces of a slab. On optimization problems the slack intervals cover 48% of the range on average, and the few rows whose slack changes side are the narrow ones (8%).

SC therefore uses the dual warm start in the solver of systems

of equations, where the certified bound is valid and nondecreasing at every iteration of every LP, and the primal warm start in the optimizer, where the first LP of each pass is still solved by the dual method. Primal steps are also the fallback when relocation is impossible because the bound indicated by a reduced cost is infinite.

4.3 Simplex iterations

We recall the bounded-variable simplex in the notation of LP_k (3), following [18]. Let B be the basis matrix, $\mathcal{B}$ and $\mathcal{N}$ the basic and nonbasic index sets, and x_N the nonbasic variables, each at one of its bounds; the basic values are $x_B = -B^{-1}\bar{A}_{\mathcal{N}} x_N$, since the right-hand side of (2) is zero. Every iteration exchanges one basic and one nonbasic variable and needs three vectors:

- the *pivot column* $\alpha_q = B^{-1}\bar{A}_{:q}$ of the entering variable z_q, obtained by solving $B\alpha_q = \bar{A}_{:q}$ (forward transformation, FTRAN);
- the *pivot row* $\alpha_r^T = e_r^T B^{-1}\bar{A}$ of the variable leaving at position r, obtained by solving $B^T\rho_r = e_r$ (backward transformation, BTRAN) and then computing $\alpha_r^T = \rho_r^T \bar{A}$;
- the step lengths: a primal step θ_p for x_B and a dual step $\theta_d = d_q/\alpha_{rq}$ for (λ, d).

The updates are then

$$x_B \leftarrow x_B - \theta_p\,\alpha_q, \qquad \lambda \leftarrow \lambda + \theta_d\,\rho_r, \qquad d \leftarrow d - \theta_d\,\alpha_r,$$

with z_q taking position r of the basis. The certificate needs λ, which is therefore always at hand.

Dual steps. In a dual iteration the basis is dual feasible and some basic variable violates its bounds. The leaving variable is chosen among the violated basic variables by the largest ratio of the squared violation to a *devex* weight [15], a cheap approximation of the norm of the corresponding row of B^{-1} that avoids the bias of the plain largest-violation rule towards badly scaled rows. The entering variable comes from the ratio test over the pivot row: among the nonbasic z_j whose reduced cost would change sign in the wrong direction, the one with the smallest $|d_j/\alpha_{rj}|$ keeps the basis dual feasible. We use the long-step (bound flipping) version of this test [20, 18]: the dual objective along the ray is piecewise linear and concave, with a breakpoint at each ratio; when a breakpoint belongs to a boxed variable, the step can go past it by moving that variable to its other bound, and the test continues while the slope remains positive. One iteration thus does the work of several short steps. Among the candidates at the chosen breakpoint we take the largest $|\alpha_{rj}|$ within a relative band (Harris' rule), which avoids tiny pivots.

Dual degeneracy. The bound problems are highly dual degenerate. With objective $\pm z_k$ and z_k basic in row r, the reduced costs after relocation are $d = \mp\alpha_r$ on the nonbasic variables: exactly zero on every column that does not meet row r. Those columns give ratio zero in the dual ratio test, so they enter first and the dual objective does not move. On some systems (Virasoro, Katsura) 80–95% of the dual pivots were degenerate. SC uses the standard remedy [18]: after relocation, each zero reduced cost is replaced by a random value in $[\varepsilon, 2\varepsilon)$, $\varepsilon = 10^{-7}$, with the sign that its current bound already satisfies. The basis stays dual feasible for the perturbed costs and the ties disappear; at the optimum of the perturbed problem the exact reduced costs are recomputed, and

the few columns that are then dual infeasible are fixed by primal steps. The multipliers, and hence the certificate, always come from the true costs. The value of ε is not critical: between 10^{-9} and 10^{-5} the results are the same, and perturbing only after a run of degenerate pivots, as LP solvers do on stalling, gives no further gain.

Primal steps. In a primal iteration the basis is primal feasible and some nonbasic variable has a reduced cost of the wrong sign; it enters (Dantzig's rule: largest $|d_j|$), and the ratio test over the pivot column α_q finds the first basic variable to reach a bound, or lets z_q move to its own opposite bound. Pivots below a stability threshold are filtered out before choosing the leaving variable; without this filter a rejected pivot was indistinguishable from optimality, and 40% of the reported optima were not dual feasible. SC takes primal steps in the primal warm start of the optimizer, and when a nonbasic variable cannot be relocated.

Tolerances and infinite bounds. The primal feasibility tolerance of z_j is $\min\{10^{-9}(1+|v|),\ \tau \operatorname{wid} \mathbf{z}_j\}$, where v is the bound being tested and $\tau = 10^{-3}$. On narrow boxes a fixed tolerance can be wider than the box itself, and the simplex then accepts as feasible a polytope that is empty. Slacks may have an infinite side; a nonbasic variable always sits at a finite bound, the other one if the bound indicated by its reduced cost is infinite, and at zero if it is free.

4.4 Two engines

The iteration above can be implemented in two ways, which differ in how B^{-1} is represented; SC has both and chooses per node by

the number of rows.

Tableau engine. On small relaxations, below the threshold given at the end of this section, the simplex keeps the whole tableau $T = B^{-1}\bar{A}$ explicitly, row by row in sparse form, together with λ, d and x_B. The pivot row and column are then read from T, with no linear system to solve, and a pivot is a Gauss–Jordan elimination: each row t with $\alpha_{tq} \neq 0$ is replaced by $T_{t:} - (\alpha_{tq}/\alpha_{rq})\, T_{r:}$, at a cost of $O(\mathrm{nnz}(\alpha_q)\,\mathrm{nnz}(\alpha_r))$. Errors accumulate in the updated rows, so the tableau must be recomputed from $\bar{A}$ from time to time; instead of a fixed period, a guard compares λ, d and x_B with their definition and triggers the recomputation only when it detects drift.

Revised engine. For larger relaxations the dense rows of T would cost $O(m(n+m))$ per pivot. The revised engine keeps instead a sparse LU factorization $B = LU$, computed by the factorization code of SoPlex, and obtains each iteration's vectors by triangular solves: FTRAN solves $Ly = \bar{A}_{:q}$ and then $U\alpha_q = y$, BTRAN solves $U^T w = e_r$ and then $L^T \rho_r = w$. Since the basis changes by one column per iteration, the factors are not recomputed but updated: in the Forrest–Tomlin update [12] the new column replaces column r of U, which breaks its triangular shape; a permutation moves it to the end, and one row elimination, stored as a sparse row transformation appended to L, restores the triangle. The update reuses the vector y of the last FTRAN, so it costs about one more solve. Updates make the factors grow, and SC refactorizes B from scratch every 50 pivots. The FTRAN of the entering column is computed once per iteration and serves three purposes: the devex weights, the Forrest–Tomlin update and the update of x_B, which is recomputed from scratch every 32 pivots to limit drift. The reduced costs and λ are updated with the pivot row as above.

A revised iteration costs one FTRAN, one BTRAN, one product $\rho_r^T \bar{A}$ and the update, all proportional to the number of nonzeros involved, instead of a pass over the tableau. Below a threshold the tableau is cheaper; above it the revised engine is. The threshold is 30 rows in the solver of systems of equations, and 15 rows in the optimizer, where the warm start is primal; both were chosen on the systems of equations and on the 139 optimization problems used for tuning (Section 6). The ablation (Table 8) shows that the tableau alone is 8–15% slower on systems of equations and 7–12% on optimization problems, and the revised engine alone 2–3% slower on optimization problems.

4.5 Certification

When an LP terminates, λ is read from the final basis: in the tableau engine it is maintained explicitly, and in the revised engine it is updated with the pivot row. The certificate $\gamma = \bar{A}^T \lambda$ is computed once per bound in outward-rounded interval arithmetic, over the rows with $\lambda_i \neq 0$, and $\mathbf{C}_k$ and $\mathbf{R}$ follow in $O(n+m)$ (Definition 2). The simplex itself never needs to be correct: a wrong pivot or an inaccurate λ yields a weaker bound, never an invalid one (Proposition 3).

5 Properties of the certificate

All statements hold for the floating-point A returned by the linearizer; rounding in the computation of γ and $\mathbf{C}_k$ is covered by outward rounding. We write e_k for the k-th unit vector of $\mathbb{R}^{n+m}$.

5.1 Validity

Proposition 3 (Validity for any multipliers)**.** *Let $\lambda \in \mathbb{R}^m$ be arbitrary. For every $z \in P(\mathbf{x})$ and every k with $0 \notin \gamma_k$, we have $z_k \in \mathbf{C}_k(\lambda, \mathbf{z})$. If $0 \notin \mathbf{R}(\lambda, \mathbf{z})$, then $P(\mathbf{x}) = \emptyset$, and by Lemma 1 the box $\mathbf{x}$ contains no solution.*

Proof. Let $\gamma = \bar{A}^T\lambda$ be the exact vector, so $\gamma \in \gamma$. For $z \in P(\mathbf{x})$, $\gamma^T z = \lambda^T \bar{A} z = 0$, hence $\gamma_k z_k = -\sum_{j \neq k} \gamma_j z_j$. Each product $\gamma_j z_j$ belongs to $\gamma_j \mathbf{z}_j$ and the sum to the interval sum; dividing by $\gamma_k \in \gamma_k$, which does not contain 0, gives $z_k \in \mathbf{C}_k(\lambda, \mathbf{z})$ by the inclusion property. For the second claim, any $z \in P(\mathbf{x})$ would give $0 = \gamma^T z \in \mathbf{R}(\lambda, \mathbf{z})$. □

No property of λ is used: it need not be optimal, feasible, or accurate. Errors in λ can only weaken the enclosure, never invalidate it. In particular, a Farkas multiplier read from the simplex when it detects primal infeasibility either passes the residual test or is simply not used: there is no "infeasible but unproved" outcome that forces the pass to stop. A warm-started simplex can, however, accumulate enough error to report infeasibility on boxes that are not empty: the ray then fails the test, the bound is merely not contracted, and the node is bisected instead of pruned. When this happens SC solves the same LP again from a cold start. In the benchmarks of Section 6 this retry changes nothing at 60 s (the search is identical in 96% of the runs), but it removes the rare long stalls in which the warm start reports such infeasibility over a whole region of the tree (Section 6.3).

Proposition 4 (Validity on sub-boxes)**.** *Let $\mathbf{x}' \subseteq \mathbf{x}$ and let $\mathbf{z}' \subseteq \mathbf{z}$ be the corresponding box of (2), built with the matrix $\bar{A}$ computed on $\mathbf{x}$. Then every solution in $\mathbf{x}'$ satisfies $z_k \in \mathbf{C}_k(\lambda, \mathbf{z}') \subseteq \mathbf{C}_k(\lambda, \mathbf{z})$.*

Proof. The linearization (1) on $\mathbf{x}$ remains valid for solutions in $\mathbf{x}'$, so $P' = \{z \in \mathbf{z}' : \bar{A}z = 0\} \subseteq P(\mathbf{x})$ contains them, and Proposition 3 applies to P'. The inclusion follows from the inclusion isotonicity of interval arithmetic. □

A certificate computed at a node can thus be reused in all its descendants, as we did with the preconditioner of [26] and Gleixner et al. with their dual-derived inequalities [13]. It stays valid, but it was optimized for the parent box and is usually weak in a child; with SC, where a fresh certificate costs one LP, we measured such reuse and it does not pay (Section 7).

5.2 Tightness

Consider the bound problem LP_k (3) with $\mathbf{z}$ bounded, and the notation of Section 4.

Lemma 5. *For any λ with $d = e_k - \bar{A}^T\lambda$ and $d_k = 0$, the exact certificate satisfies $\gamma_k = 1$ and*

$$\inf \mathbf{C}_k(\lambda, \mathbf{z}) \;=\; \sum_{j \neq k} \min_{z_j \in \mathbf{z}_j} d_j z_j \;=\; \min_{z \in \mathbf{z}} d^T z \;=: g(\lambda),$$

the value of the Lagrangian dual function of (3) at λ.

Proof. $\gamma = \bar{A}^T\lambda = e_k - d$, so $\gamma_k = 1 - d_k = 1$ and $\gamma_j = -d_j$ for $j \neq k$. Then $\mathbf{C}_k = \sum_{j \neq k} d_j \mathbf{z}_j$, whose infimum is the sum of the minima of the independent terms. The term $j = k$ of $d^T z$ vanishes. Finally, the Lagrangian of (3) relaxing $\bar{A}z = 0$ with multiplier λ is $z_k - \lambda^T \bar{A} z = d^T z$, minimized over the box $\mathbf{z}$. □

Proposition 6 (Exactness at an optimal basis)**.** *Let B be an optimal basis of (3) with multipliers λ^*. If z_k is basic, then in exact*

arithmetic $\inf \mathbf{C}_k(\lambda^*, \mathbf{z})$ *equals the optimal value of* LP_k. *If* z_k *is nonbasic at its lower bound, the optimal value is* $\inf \mathbf{z}_k$ *and there is nothing to contract.*

Proof. If z_k is basic, $d_k = 0$ and Lemma 5 gives $\inf \mathbf{C}_k = g(\lambda^*)$. By weak duality $g(\lambda^*)$ is a lower bound of the optimum. The basic solution $\bar{z}$ places each nonbasic variable at the bound that minimizes $d_j z_j$ (optimality signs) and has $d_j = 0$ on basic variables, so $d^T \bar{z} = g(\lambda^*)$; since $\bar{z} \in P(\mathbf{x})$, $\bar{z}_k = e_k^T \bar{z} = d^T \bar{z} + \lambda^{*T} \bar{A} \bar{z} = g(\lambda^*)$. If z_k is nonbasic at its lower bound, $\bar{z}_k = \inf \mathbf{z}_k$ is feasible and optimal. □

The remaining case, z_k nonbasic at its upper bound in an optimal basis of the minimization, means that $P(\mathbf{x})$ lies in the hyperplane $z_k = \sup \mathbf{z}_k$. Then $d_k \le 0$, so $\gamma_k = 1 - d_k \ge 1$, and the same computation as in Lemma 5, divided by γ_k, shows that the certificate is exact as well.

In floating point, the computed enclosure exceeds the LP bound by at most $\sum_j \mathrm{wid}(\gamma_j)\ \mathrm{mag}(\mathbf{z}_j)$ plus the effect of the inaccuracy of λ, the same slack as in the Neumaier–Shcherbina bound [23]. On identical boxes, SC attains the LP hull in 96% of the bounds where it contracts more than the standard contractor (Section 6).

Remark 7 (Relation to Neumaier–Shcherbina and Domes–Neumaier). Written back in the variables x, $\inf \mathbf{C}_k(\lambda, \mathbf{z})$ is the Neumaier–Shcherbina safe lower bound of $\min x_k$ subject to $Ax \in \mathbf{b}$, $x \in \mathbf{x}$, with approximate dual λ [23], and the row-wise solution of the preconditioned system in the linear bounding of Domes and Neumaier [10, §5]. In the homogeneous form, Proposition 6 is our earlier result [26, Prop. 1] that the dual solution of the bound LP is an optimal preconditioner for the Gauss–Seidel projection, with the vector $\lambda^T \bar{A}$ playing the role of a preconditioned row. The certificate is therefore the known one; Propositions 8 and 9 are what

make it cheap and always available inside the simplex.

5.3 Dual feasibility for free

Proposition 8 (Dual-feasible start for every bound)**.** *The slack basis $B = -I_m$ is dual feasible for* LP_k*, for every structural variable $k \le n$ with* $\inf \mathbf{z}_k$ *finite. More generally, if every $\mathbf{z}_j$ is bounded, any basis becomes dual feasible after placing each nonbasic z_j at its lower bound if $d_j > 0$ and at its upper bound if $d_j < 0$.*

Proof. With the slack basis, $c_B = 0$ for $k \le n$, so $\lambda = 0$ and $d = e_k$: the only nonzero reduced cost is $d_k = 1$, and z_k at its lower bound satisfies the sign condition. In general, dual feasibility is a condition on the signs of d_j for nonbasic variables relative to the bound at which they sit; d does not depend on that placement, so the relocation enforces the condition whenever the required bound is finite. □

When the objective changes from one bound to the next, or the box is contracted between two bounds, relocation restores dual feasibility without a phase 1. That a boxed LP needs no dual phase 1 is a standard observation [18]; what we use is that it holds across the $2n$ objectives and the contractions of the box, which is what allows one tableau to be shared by the $2n$ problems. The case of a box change was exploited by Yamamura and coworkers [29, 30] to resume an LP emptiness test after a bisection; Proposition 8 extends it to a change of objective, which is what the $2n$ bound problems need. Relocation needs the bound indicated by the sign of the reduced cost to be finite. Slacks of one-sided rows, and variables with an unbounded domain, may not have it. In the implementation such a nonbasic variable is placed on its finite bound, which leaves its reduced cost with the wrong sign, or at zero if it

is free. The iteration loop removes these dual infeasibilities with primal steps, and uses dual steps while some basic variable is out of its bounds; the revised engine of the solver of systems of equations takes no primal steps, and an LP that ends with such a variable gives a valid, possibly weaker bound. The certificate is unaffected, since it is valid for any λ; only the monotonicity of Proposition 9 is lost in such an LP until the basis is dual feasible again.

5.4 An anytime certified bound under the dual method

Proposition 9 (Monotone certified bound)**.** *Run the dual simplex method on* (3) *from a dual-feasible basis, with the standard or the long-step ratio test. At every iteration t in which z_k is basic,* $\inf \mathbf{C}_k(\lambda_t, \mathbf{z})$ *is a valid lower bound on z_k over $P(\mathbf{x})$, and, in exact arithmetic, these bounds are nondecreasing in t.*

Proof. Validity is Proposition 3. With z_k basic, $d_k = 0$ and Lemma 5 identifies the bound with the dual objective $g(\lambda_t)$. Every iterate of the dual simplex is dual feasible, and each iteration moves λ along a dual ascent direction with a step chosen by the ratio test; the long-step test maximizes the piecewise-linear dual function along that direction [20]. Hence $g(\lambda_t)$ is nondecreasing, strictly so on nondegenerate iterations. □

The certificate can thus be read at any point of the dual method, and an LP can be truncated with a valid bound. We measured such truncation and it does not pay (Section 7). In the solver of systems of equations, where SC uses the dual warm start (Section 4), the property holds in every LP of the pass; in the optimizer it holds in the first LP of each pass. With the cost perturbation of Section 4, what is nondecreasing is the dual objective of the perturbed problem; the certified bound, computed with the true costs, differs from

it by at most $2\varepsilon \sum_j \operatorname{mag} \mathbf{z}_j$, where $\operatorname{mag} \mathbf{z}_j = \max(|\inf \mathbf{z}_j|, |\sup \mathbf{z}_j|)$: about 10^{-7} times the magnitude of the box.

Cost. Certifying a bound costs the same in SC as the Neumaier–Shcherbina post-processing. Both compute a product of the transposed constraint matrix by the multipliers, restricted to the rows with $\lambda_i \neq 0$, plus $O(n+m)$ operations: $\gamma = \bar{A}^T\lambda$ and $\mathbf{C}_k$ in SC, and the residual $A^T\lambda - c$ and the bound $\lambda^T\mathbf{b} - (A^T\lambda - c)^T\mathbf{x}$ in Neumaier–Shcherbina. On three optimization problems (ex6_1_3, ex7_2_3, chem) the SC certificate costs 1.3–3.1 μs per LP, and the post-processing 1.9–3.0 μs plus 1.2–1.5 μs to extract the solution from the external solver. The difference between the two approaches is therefore not the cost of a certificate but where λ comes from, which is what Section 6 isolates.

6 Experimental evaluation

A controlled comparison. To measure the contribution of the simplex, and not of the rest of the contractor, we compare two engines inside the *same* contractor. Both run the same pass over the $2n$ bounds, with the same skipping rule, the same certificate and the same policy when an LP yields no certificate: that bound is not contracted and the pass continues. They differ only in where the multipliers come from:

- *SC*: the multipliers of the final basis of the self-certifying simplex;
- *BB* (black box): SoPlex 4.0.2 solves each LP on the same relaxation (warm-started from the previous bound, as in the polytope-hull contractor), and its dual solution, or its Farkas

ray if it reports infeasibility, is passed to the same interval certificate $\gamma = \bar{A}^T\lambda$, which is the Neumaier–Shcherbina bound written in the homogeneous form.

The homogeneous form is therefore not what separates the two engines: the certificate, its evaluation over the box and its validity on sub-boxes are common to both, and SoPlex itself solves ranged rows in the same form, with one bounded logical variable per row. What differs is a simplex designed for that certificate: the free dual-feasible start, one basis shared by the $2n$ bounds, and the warm start between bounds. The black box was given its best configuration. It runs on its own SoPlex instance holding the same rows and sides, with the box only as column bounds: the LP interface of Ibex also stores the n variable bounds as rows, which makes each LP of SoPlex 16% more expensive on small relaxations and up to twice as large on square systems (with them, the comparison of Table 3 was 6–11 points more favorable to SC on systems of equations and 2 on optimization problems). Rows that are free for SoPlex (both sides infinite) are passed to it empty: such a row constrains nothing, but in the affine relaxation of Brown the row of the product equation has coefficients of order 10^{108}, on which SoPlex 4.0.2 and 7.1.6 alike fail in almost every LP. We also tried SoPlex 7.1.6 as the black box: on a sample of eight problems it explores the same cells or fewer per second and is 1.2–1.9 times slower than 4.0.2, also with presolving and scaling off, so we keep 4.0.2; the revised engine of SC, which uses the LU factorization of SoPlex, gives the same search trees with the LU of 7.1.6 and is 0–14% slower with it. Its feasibility and optimality tolerance matters: 10^{-9} makes SoPlex report spurious infeasibility on relaxations with nearly-equal rows, while 10^{-6} weakens the dual on optimization problems; we use 10^{-6} for systems of equations and 10^{-9} for optimization, the best value in each case. Turning off presolving or scaling, or forcing

the row or column representation, did not help; forcing the primal algorithm reduced the spurious infeasibility but made the overall results worse. The bound order matters for the black box too: on systems of equations, the nearest bound first makes its warm starts cheaper and saves 3–5% of CPU time with the same cells and the same runs finished, and 4–7% on optimization problems. The black box therefore uses the nearest bound first, and SC its own order. We also compare both engines with the farthest bound first.

Setting. Contractor and solvers are implemented in Ibex with the affine relaxation [25]. The affine linearizer writes each equation as two one-sided rows with the same normal. In the homogeneous form they are a single row whose slack has the interval of both sides, so we merge them, for both engines. This halves m on square systems, and it matters for the black box: two half-spaces with the same normal and right-hand sides 10^{-14} apart are a source of spurious infeasibility for SoPlex (without merging, the black box explores 20 times more cells on CountercurrentReactors2-9). For systems of equations we use the solver composition of Ibex (HC4, ACID, interval Newton when the system is square, and the linear contractor within a fixed point with HC4) with the LSmear bisector, on the 242 systems of the Ibex solver benchmark that some configuration solves within 3600 s (22 are excluded), 2 seeds, 60 s per run. For optimization we use IbexOpt on 240 problems, those of the Ibex optimization benchmark that some configuration solves within 3600 s (59 are excluded), 4 seeds, 60 s per run. The parameters of SC were chosen on 139 of them, except the bound order of the optimizer, chosen on all 240; on the other 101, which we added afterwards, SC is 17–20% faster, against 18–20% on the 139. Three problems (alkyl, alkylbis, bearing) whose search trees change by two orders of magnitude between seeds of the same configuration

are excluded from the means. We report geometric means of ratios over the pairs of runs in which both methods finish, for runs whose reference time exceeds 0.1, 1 and 5 s, and the number of runs that finish within the time limit. All runs were made on a machine with two Intel Xeon E5-2620 v4 processors (2.1 GHz, 16 cores) and 64 GB of memory, under Linux 6.17, eight runs at a time and a memory limit of 6 GB per run; the code is compiled with GCC 13.3 (`-O3`) against Ibex 2.8.9 with the GAOL 4.2.0 interval library and SoPlex 4.0.2. CPU times are process times. **[TODO:** publicar el codigo y los scripts**]**

6.1 Rigor

SC is rigorous by Proposition 3; we also checked it empirically against the LP hull. On identical boxes and with the same polytope, for every bound where SC contracts more than the polytope-hull contractor of Ibex, we solved the bound LP of the polytope over the input box with a fresh instance of SoPlex, used as an oracle. No valid contraction can go beyond this hull. Over 3 090 such bounds in four systems (Table 2), SC attained the hull in 96% of the cases and went beyond it in 27, all on boxes narrower than $2 \cdot 10^{-5}$ and by at most $3.4 \cdot 10^{-9}$ in absolute value ($2.8 \cdot 10^{-10}$ relative to the bound): below the feasibility tolerance of the oracle, 10^{-9}, whose own optimal point lay outside the input box in 21 of these 27 bounds. No bound went beyond the hull by more than that tolerance.

6.2 Main comparison

Table 3 compares SC with the black box. On optimization problems, whose relaxations are small, SC is 17–20% faster with similar search trees (18–19% with the farthest bound first in both). On sys-

Table 2: Bounds where SC contracts more than the polytope-hull contractor of Ibex on the same box, compared with the LP hull of the polytope over the input box computed by SoPlex. "Beyond" counts bounds beyond the hull, all within the 10^{-9} tolerance of the oracle.

system	bounds	SC at hull	PH at hull	SC beyond
Bratu-0060	892	870	40	0
Discrete-Integralf2-13	1 048	1 025	59	22
Geneig	860	803	5	0
Bellido	290	282	21	5
total	3 090	2 980	125	27

tems of equations SC is 9–10% faster (12% with the farthest bound first in both) and explores 2–5% fewer cells; it finishes 364 runs against 362. The largest gains are on Geneig-icse (×0.33), where the black box leaves many LPs without an optimum (Section 6.5), and DiscreteBoundary-0200 (×0.29). On Brown-18 to 30 both engines explore the same cells and SC is 14% faster. On the largest relaxations (Bratu-0120 to 0200, $m = n$ up to 200 after merging) the black box with the nearest bound first is slightly faster (×1.03 for SC on the family; ×0.81 with the farthest bound first in both). SC loses by more than 5% on three systems (Bratu-0175 ×1.27, Katsurabis-14 ×1.23, Bratu-0180 ×1.13); with the same bound order it loses only on Katsurabis-14 (×1.33).

Figure 1 shows the same comparison as performance profiles [9] over the runs above 0.1 s that at least one engine finishes: SC is the faster engine in 85% of the runs on systems of equations and in 91% on optimization problems; within a factor of 2 the black box reaches 98% of the runs in both, and it does not finish 0.8% and 1.2% of them, all of which SC finishes; SC fails to finish one run

Table 3: SC against the best black-box (BB) configuration, inside the same contractor. Ratios SC/BB (cells and CPU) for runs whose reference time exceeds 0.1, 1 and 5 s, and runs finished. Best of each: SC with its own bound order, the black box with the nearest bound first. The last rows use the farthest bound first in both engines.

			CPU			
benchmark	configuration	cells	> 0.1 s	> 1 s	> 5 s	finished
optimization	best of each	1.00–1.02	0.80	0.80	0.83	216 / 216
systems	best of each	0.95–0.98	0.90	0.91	0.90	364 / 362
optimization	same bound order	0.99–1.01	0.81	0.82	0.82	217 / 216
systems	same bound order	0.96–0.98	0.89	0.88	0.88	363 / 362

that the black box finishes (alkyl, seed 2, one of the three problems excluded from the means).

6.3 Long runs

The 60 s limit leaves out the hardest instances. We reran with a limit of 3600 s the instances on which some configuration reached the 60 s limit (48 systems with 2 seeds, 24 optimization problems with 4 seeds), each engine with its own bound order, and on systems also with the farthest bound first in both. Both engines re-solve from a cold start when a reported infeasibility cannot be certified (Section 5); for the black box this also covers infeasibility reported without a ray. Table 4 gives the results.

On optimization problems the gain of Table 3 carries over. SC finishes three problems (ex6_2_13, hs101, hs112) in 3 000 to 3 100 s that the black box does not finish within 3 600 s; with 7 200 s it finishes them in 4 000 to 4 480 s with the same cells, and

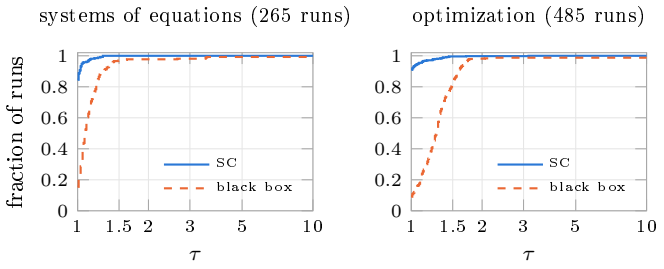


Figure 1: Performance profiles of CPU time, SC (own bound order) against the black box, over the runs above 0.1 s that at least one engine finishes: fraction of runs solved within a factor τ of the faster engine.

these runs enter the means. Without bearing, whose search tree changes by two orders of magnitude between seeds (Section 7), SC is 22% faster with 2% fewer cells; the largest loss is ex5_4_4 ($\times$1.18). On long runs of systems of equations the two engines are close: SC is 3% faster with its own bound order and 5% faster with the farthest bound first in both, with the same cells, and finishes one more run (CountercurrentReactors2-26, in 257 s). The mean with each engine's own order is dominated by one run, CountercurrentReactors2-28 with seed 1 ($\times$4.2); without it SC is 5% faster with 1% fewer cells. That run is not caused by the linear contractor: the adaptive tuning of ACID, which every 1 000 calls re-estimates how many variables to shave from a window of 50 calls, settles on shaving none for a long stretch of the search, and boxes that ACID alone would empty are bisected instead (with a different seed, or without the retry, the same instance finishes in 320–370 s).

Table 4: SC against the black box with a limit of 3600 s, each engine with its own bound order, and on systems also with the farthest bound first in both. Geometric means of the ratios SC/BB over the runs both engines finish, and runs finished.

benchmark	instances	runs	cells	CPU	finished
systems	48	96	1.02	0.97	94 / 93
systems, same bound order	48	96	1.00	0.95	94 / 94
optimization	24	96	0.95	0.77	96 / 84

6.4 Cost of a pass

Table 3 mixes two effects: how many cells are explored and what each pass of the contractor costs. To measure the polytope-hull contractor alone, without the noise of the search, we built a bank of passes: for 71 instances, one per family on systems of equations and 35 optimization problems spread over the range of running times, we recorded the input boxes of 1 000 calls of the contractor along the search of SC, and replayed each call outside the search with both engines on the same LPs. Both engines then contract every box to the same result (mean contraction 0.166 against 0.166 on systems and 0.273 against 0.273 on optimization problems, with the same boxes emptied), and only their cost differs (Table 5). A pass of SC costs 0.55 times that of SoPlex on systems of equations and 0.47 times on optimization problems; it is cheaper on 33 of the 36 systems and on all 35 optimization problems. The three exceptions are DiscreteBoundary-0200, Katsurabis-14 and Bratu-0120, discussed in Section 7.

To attribute this difference we used the same bank in two directions (Table 6). On one side we added to SC, at every LP and on separate vectors, so that its contraction does not change, each

Table 5: Polytope-hull passes replayed outside the search (1 000 recorded calls per instance): time per pass of SC over that of SoPlex on the same LPs (geometric mean and medians), instances where SC is cheaper, and mean contraction of the perimeter with each engine.

benchmark	instances	SC/BB	median (μs)	SC cheaper	contraction (SC / BB)
systems	36	0.55	215 / 412	33	0.166 / 0.166
optimization	35	0.47	121 / 261	35	0.273 / 0.273

part of the work that SoPlex does in a call; on the other we removed from SoPlex the parts that can be removed from outside (with a patched copy for the verification and the refactorization at optimality). In logarithmic scale the shares add up. Refactorizing the basis at the start of each LP would account for 41–45% of the gap, verifying the solution for 16–19% (10–18% when it is removed from SoPlex, the one part measured both ways), and presolving each new relaxation, which only SoPlex does, for 22–23%; recomputing the vectors at the start of a solve and copying the solution add little, and turning off the scaling of SoPlex makes it slower on systems of equations. With everything added, a pass of SC would cost 0.85 and 0.76 times that of SoPlex: these parts explain 73% and 65% of the gap, and the rest is the general machinery of a call (setting up and storing vectors, unscaling, the interface).

The same comparison can be made inside the search. We ran both engines on the same LPs at every node: the black box first runs the pass on a copy of the box, and SC then runs it on the same box and polytope, with the search following SC. Figure 2 compares the time of a full pass over the bounds, including ordering, skipping and certification, which are common to both. A pass of SC costs 0.62 times that of the black box on systems of equations (geometric mean

Table 6: Attribution of the gap of Table 5, as the share of the logarithm of the time ratio. Left: work of SoPlex added to SC at every LP, cumulatively. Right: work removed from SoPlex, cumulatively. The time per pass relative to SoPlex is given after each step (34 optimization problems: one emulation failed on launch).

added to SC	systems	optim.	removed from SoPlex	systems	optim.
(SC)	0.55	0.46	(SoPlex)	1.00	1.00
refactorize per LP	45%	41%	verification	18%	10%
recompute vectors	7%	6%	refactorization at optimum	1%	3%
verify the solution	19%	16%	presolve	22%	23%
copy the solution	2%	1%	scaling	−22%	4%
result	0.85	0.76	result	0.90	0.74

over the 183 systems with at least 20 nodes and 100 LPs; median 417 against 682 μs) and 0.44 times on optimization problems (175 problems; median 114 against 254 μs), and is cheaper on 85% and 99% of the instances. Each engine uses its own bound order, as in Table 3.

The two other panels split this ratio into pivots per pass and time per LP. SC does not pivot less: it makes 33% more pivots than the black box on systems of equations (3.0 against 2.0 per LP) and 37% more on optimization problems (2.4 against 1.8). The whole gain is in the cost of an LP ($\times$0.68 and $\times$0.41; medians 25.1 against 38.2 and 12.3 against 30.2 μs). With two or three pivots per LP, the black box pays mostly for what a call does besides pivoting. On ex2_1_9 an LP of the black box takes 24 μs against 8 μs for SC, with about two pivots per LP in both. Counting instructions on the same LPs, the pivots themselves cost about the same in both engines (9 and 10 thousand instructions per LP), while SoPlex spends 16% of a call rebuilding its vectors at the start of every solve, 20% storing, unscaling and verifying the solution, 11% presolving

each new relaxation and 7% refactorizing; an LP of the black box takes 74 thousand instructions, against 26 thousand for SC. No single part dominates: suppressing the refactorizations that SoPlex makes before declaring optimality (from 1.2 to 0.25 per LP) did not change the time of an LP. SC solves the $2n$ bounds of a pass on one tableau and only replaces the objective row and the bounds that changed. The cascade of Table 6 confirms this attribution on the whole bank of passes. On large relaxations or LPs with many pivots the comparison reverses: on Katsurabis-14 and Bratu-0060 an LP of SoPlex is cheaper than one of SC (105 against 203 and 80 against 110 μs), because it pivots less (Section 7); these are the instances where SC loses. Where the LPs need many pivots this fixed cost is amortized: on launch (9 to 11 pivots per LP) an LP costs about the same in both engines (78 against 81 μs). Since the pass is only part of the cost of a node, with the linearization, propagation, ACID and interval Newton, its lower cost yields the 9–20% of Table 3.

6.5 Where the black box loses bounds

With the same certificate, the two engines can only differ in the multipliers they produce. We ran both engines on the same input boxes and the same polytope at every node (Table 7). Two mechanisms appear.

1. *No optimum.* On some relaxations SoPlex ends without an optimal basis, and the bound is lost. On Geneig-icse SoPlex reaches the iteration limit or reports an infeasibility it cannot prove on 379 LPs, against 70 for SC. (On Brown, before free rows were passed to it empty, SoPlex failed with an internal exception in 90% of the LPs; SC solves them all.) SC has no such outcome by design: any basis certifies, and an LP that stops early yields a weaker bound, not a lost one. With the

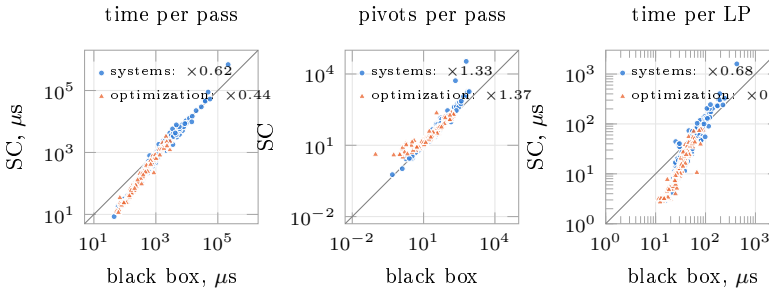


Figure 2: Cost of a polytope-hull pass with both engines on the same LPs (same boxes and polytope at every node), one point per instance; points below the diagonal favour SC. Left: time per pass. Middle: pivots per pass. Right: time per LP.

twin rows unmerged this mechanism was dominant (a third of the LPs of the black box on Geneig-icse); merging them removes most of it.

2. *Dual accuracy.* When both engines reach an optimum, the certificate built from the multipliers of SC is often slightly tighter. On Geneig-icse, SC contracts more in 128 of 1 710 nodes and the black box in 9, and on CountercurrentReactors2-9 in 66 against 2; on Bellido and Brown-18 the two engines tie (4 against 4, 1 against 1). The difference per node is small but compounds along the search; we have not isolated its cause.

Table 7: Both engines on the same input boxes and the same polytope. Mean relative contraction of the perimeter, nodes where each engine contracts more (nodes that both empty count as ties), and LPs without a usable certificate (black box with tolerance 10^{-6}).

	contraction		nodes won		LPs not optimal	
system	SC	BB	SC	BB	SC	BB
Geneig-icse	0.323	0.323	128	9	70	379
CountercurrentReactors2-9	0.046	0.047	66	2	110	0
Brown-18	0.314	0.305	1	1	2	0
Bellido	0.281	0.281	4	4	152	0

6.6 Ablation

Table 8 changes one component of SC at a time. The pruning changes little: cell ratios are within 3.5% in every row. The components differ in cost, mostly on systems of equations, whose relaxations are larger. Merging the twin rows is the largest saving (without it, CPU time $\times$1.26–1.42 and eight fewer runs finished), followed by the hybrid engine (with the tableau alone, $\times$1.08–1.15 and seven fewer runs on systems of equations, $\times$1.07–1.12 on optimization problems). The revised engine alone is about as fast on systems of equations, but 2–3% slower on the small relaxations of optimization problems; the threshold keeps the best of both. Among the pivoting rules, the long-step ratio test gives 7–10% on systems of equations, the cost perturbation 3–4% (and one more run finished), devex 1–2%, and the box-scaled tolerance is neutral in time: it is there for correctness on narrow boxes (Section 4). On optimization problems these rules are within 2%; the perturbation, applied when the dual warm start relocates the nonbasic variables, does not act there, and the search is identical without it. Each pro-

Table 8: Ablation: each row changes one component of SC. Ratios against full SC: cells and CPU on systems of equations for runs above 0.1, 1 and 5 s, runs finished out of 484 (full SC: 364), CPU on optimization problems above 0.1 s and problems solved in all seeds (full SC: 216). The warm-start row gives each program the warm start of the other: primal on systems of equations, dual on optimization problems; the order row gives each the order of the other: nearest bound first on systems of equations, the count read from the tableau on optimization problems.

	systems of equations					optimization	
configuration	cells	> 0.1 s	> 1 s	> 5 s	fin.	CPU	solved
without cost perturbation	1.02	1.03	1.04	1.04	363	1.01	215
warm start of the other program	1.01	1.05	1.05	1.04	363	1.09	216
without long-step ratio test	1.01	1.10	1.08	1.07	362	1.01	217
without devex	1.01	1.01	1.02	1.00	363	1.00	217
without box-scaled tolerance	1.00	1.01	1.00	1.00	364	1.00	218
tableau engine only	1.02	1.15	1.14	1.08	357	1.07	216
revised engine only	1.00	0.99	1.00	0.99	364	1.02	216
without merging twin rows	1.02	1.42	1.34	1.26	356	1.02	216
farthest-bound order	1.00	1.02	1.02	1.01	363	1.05	217
order of the other program	1.02	1.02	1.02	1.01	362	1.02	217

gram loses with the warm start of the other (Section 4): the primal warm start is 4–5% slower on systems of equations, with one fewer run finished, and the dual warm start 9–15% slower on optimization problems. The bound order follows the warm start: the count read from the tableau saves 1–2% over the farthest bound first on systems of equations, where the nearest bound first is 1–2% slower; on optimization problems the nearest bound first saves 4–5% over the farthest and 2–3% over the count.

Relation to the polytope-hull contractor of Ibex. Against the contractor distributed with Ibex, SC explores about half the cells on systems of equations. Most of that gap comes from the contractor and not from the engine: when the LP solver reports infeasibility without a verifiable ray, the contractor of Ibex abandons the pass, and the node is not pruned. The black-box engine above, inside our contractor, recovers most of it.

7 Discussion

Why SC wins. SC does not contract with a better polytope: both engines solve the same relaxation and feed the same certificate. Its advantage has two sources. The first is cost: a pass over the $2n$ bounds costs 0.62 times that of the black box on systems of equations and 0.44 times on optimization problems (Figure 2). SC does not pivot less: on the same sequence of LPs, SoPlex needs as many pivots or fewer (on ex2_1_9, 1.6 per LP against 2.0), yet the solve of an LP of SC costs a quarter (5 against 20 μs; 8 against 24 with ordering and certification, as in Figure 2). Each LP is cheaper because the $2n$ bounds share one tableau and the restart needs no phase 1, so that an LP costs little beyond its pivots, while each call to the black box pays for setting up and tearing down a general-purpose solve. The second is robustness: where the black box ends without an optimum, the bound is lost, while the multipliers of SC still certify it (Table 7); this is where the largest gain is (Geneigicse). The slightly tighter certificates of SC on the remaining LPs compound along the search but are a second-order effect.

Where it loses. SC loses where the LPs need many pivots and SoPlex needs fewer of them. Replaying the same LPs with both engines shows why: SoPlex ends each LP at a vertex where most

structural variables sit at a bound of the box, so that the variable of the next bound is usually nonbasic and that LP takes zero or one pivot; SC ends at vertices where many structural variables are basic and many rows are tight, and starts the next LP about ten infeasibilities away. On Katsurabis-14, with the same sequence of LPs, SC needs 10.4 pivots per LP against 1.2 (181 against 34 μs per solve; with each engine on its own pass, 203 against 105 μs per LP), with about 22 basic structural variables against 6.5 in SoPlex; changing the warm start, the pricing, the perturbation or the engine of SC does not change this. The third exception of Table 5, DiscreteBoundary-0200, is of another kind: on the same sequence of LPs both engines need the same pivots (12.4 against 12.7 per LP), and the difference comes from the LPs that each engine solves in its own pass, with its own order and skipped bounds. On systems of equations, SC is slower than the black box with the nearest bound first by more than 5% on three systems. On the largest relaxations its nodes cost more and it explores slightly more cells: Bratu-0175 ($\times$1.27, 55 cells against 51) and Bratu-0180 ($\times$1.13, 51 against 47); Katsurabis-14 ($\times$1.23) explores the same cells with costlier nodes. On optimization problems the largest losses are hhfair1 ($\times$1.26, 1 596–2 412 cells against 1 600–1 704) and ex7_2_3 ($\times$1.17), whose search tree varies by a factor of 4.5 between seeds for SC (17 902–81 522 cells; 16 124–23 706 for the black box); himmel16, the largest loss with the order by the count, is now even ($\times$1.03). With the farthest bound first in both engines, SC loses only on Katsurabis-14 ($\times$1.33).

What did not pay. We also measured on the whole benchmark several ideas that use the certificate or the tableau beyond the pass, and none paid. Reusing certificates of an ancestor box keeps them valid, by Proposition 3, but not strong: tested on the slices of

ACID, the multipliers of the node itself almost never prune a slice, and those of an ancestor prune slices that the linear contractor of the node prunes anyway. Truncating the LPs with the anytime bound of Proposition 9 saves pivots ($\times 0.84$) but weakens the contraction (cells $\times 1.39$–1.60). Cuts at the LP optima reduce the cells ($\times 0.80$–0.88) at a higher cost (CPU $\times 1.18$–1.31). A Lagrangian bound improved by monotonicity tightens the LP bound in most applications, but costs more than it saves (CPU $\times 1.08$ to $\times 3.5$). Inheriting the basis between nodes saves pivots ($\times 0.78$–0.96) but rebuilding the tableau costs as much. Choosing the warm start of each node by the side changes observed in the previous LPs does not improve on the choice per program. The pattern is that doing more work inside a node does not change the search tree; what changed it was what is computed: the merged relaxation, the stability filter of the ratio test, the choice of the warm start.

Limits. SC removes the external LP solver from the contractor, not from the optimizer: the upper-bounding of IbexOpt still solves its LPs with SoPlex, in both engines. On problems with very narrow equalities the upper-bounding becomes the bottleneck: on bearing, whose eight equalities are relaxed to $|h| \leq 10^{-8}$, the lower bound of SC is good but feasible points are found only in boxes of width 10^{-4} to 10^{-3}, and the search tree changes by two orders of magnitude between seeds; we excluded it, with alkyl and alkylbis, from the means. Relaxations of several hundred rows are still expensive for both engines: the systems of 400 to 1 000 variables of the benchmark exceed the limits of 60 s and 6 GB with either. The cheaper pass of SC comes from the cost of a call to SoPlex 4.0.2, whose LPs here need two or three pivots, not from pivoting less; other LP solvers have other per-call costs, which we did not measure. Making SC end its LPs at vertices of the box, as SoPlex

does, would remove most of its losses; a tie-break toward the corners of the box inside the optimal face did not, and neither did a short walk inside the optimal face at the end of each LP (up to ten primal steps that keep the optimum and push the basic structural variables toward a bound: the optimal face leaves them almost no room, and the extra pivots are not recovered). Why the dual and primal paths of SoPlex end at those vertices remains open.

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